REVISION NOTES

Automatic Warning System (AWS)

RS522 Issue 3 — December 2015

ℹ *All content sourced exclusively from RS522-Iss-3.pdf (AWS and TPWS Handbook, Issue 3, December 2015) — AWS section only.*

Automatic Warning System (AWS)

1.1 General Information

Background

The Automatic Warning System (AWS) has been implemented as the national warning system on the UK main line passenger railway network since the 1950s.

RS522 §1.1.1 *Background*

Purpose of AWS

The original concept of AWS was to provide the driver with an audible and visual indication of whether the distant signal was clear or at caution.

Should the driver fail to respond to a warning indication, an emergency brake application will be initiated.

Since the introduction of multi-aspect signalling, the majority of signals are fitted with AWS.

> ⚠ **AWS does not relieve the driver of the responsibility of observing and obeying lineside signals and indicators.**

RS522 §1.1.2 *The purpose of AWS*

Provision of AWS

AWS consists of track and train equipment. The track equipment consists of an AWS magnet that is normally provided 180 metres (approximately 200 yards) on the approach to a signal.

The AWS magnet may be positioned at a greater distance from the signal on high-speed lines or at a lesser distance from the signal on lower speed and platform lines.

This system works by the train detecting sequences and polarities of magnetic fields passing between the track equipment and the train equipment via a receiver under the train.

> ℹ *At through stations where the permitted speed is 30 mph or less and the layout is complex, AWS track equipment need not be provided. Where this occurs, these are called AWS gap areas.*

AWS magnets are not provided at semaphore stop signals. Where a distant signal is mounted on the same post as a semaphore stop signal then AWS is provided for the distant signal.

Where a line is not fitted with AWS, this is shown in the Sectional Appendix.

Where a reduction in permissible speed is provided with a warning indicator (i.e. the permissible speed on the approach is 60 mph or more and the reduction in the permissible speed is at least one third) an AWS permanent magnet is provided 180 metres (approximately 200 yards) on the approach to the warning indicator. These are sometimes referred to as 'Morpeth magnets'.

AWS magnets are also used to alert the driver to:

- Level crossing warning boards or indicators.
- Temporary speed restriction warning boards.
- Emergency speed restriction warning boards and emergency indicators.

RS522 §1.1.3 *Provision of AWS*

1.2 Track Equipment

The AWS track equipment comprises various components mounted in the centre of the four-foot.

Magnet Type	Function
Permanent magnet	The train will first encounter a permanent magnet. Where an electromagnet is either not provided or not energised, the AWS gives a warning indication to the driver.
Electromagnet	An energised electromagnet, when presented after the permanent magnet, gives the driver a 'clear' indication when approaching a green signal or a semaphore distant signal showing 'clear'.
Suppressor magnet	Used to suppress permanent magnets when they are not required to apply to a train movement (e.g. magnets applicable to the opposite direction on a single or bi-directional line).
Depot test magnet	A permanent magnet used to test the operation of a train's AWS equipment. May be provided at the exit of certain maintenance depots.
Portable magnet	Provided to give a warning to the driver on the approach to temporary and emergency speed restrictions.

The track equipment layout: permanent magnet first (in direction of travel), followed by the electromagnet.

RS522 §1.2 *Track equipment*

1.3 Train Equipment

The following equipment is provided on each fitted traction unit.

AWS Receiver

The AWS receiver is located under a traction unit and detects the sequences and polarities of magnetic fields from the AWS track magnets.

RS522 §1.3 *AWS receiver*

AWS Audible Indicator

The audible indicator gives a warning or a clear indication that is distinguishable from all other audible cab indications:

- a clear indication (bell or electronic equivalent), or
- a warning indication (horn or electronic equivalent).

RS522 §1.3 *AWS audible indicator*

AWS Visual Indicators

Visual Indication	Meaning
Black (sunflower fully black)	Advises the driver that the associated signal is showing a green aspect or 'all clear'. Also advises that the audible warning has NOT been acknowledged — if not acknowledged, brakes will be applied.
Yellow and black (sunflower yellow/black)	Advises the driver that a warning indication has been acknowledged.

RS522 §1.3 *AWS visual indicators*

AWS/TPWS Acknowledgement Button

The AWS/TPWS acknowledgement button is used to acknowledge an AWS audible warning.

> ⚠ **If an AWS audible warning is not acknowledged within two to three seconds an emergency brake application will occur.**

RS522 §1.3 *AWS/TPWS acknowledgement button*

AWS Isolation/Fault Indicator

Some traction units are fitted with a visual indicator to advise the driver of a fault with the AWS, and when the AWS has been isolated. The yellow isolation/fault indicator gives three indications:

State	Meaning
Off	AWS state is normal.
Flashing	A fault has been detected in the train AWS equipment.
On (steady)	The train AWS equipment has been isolated.

RS522 §1.3 *AWS isolation/fault indicator*

1.4 AWS Indications and Their Meanings

Warning Indication

The driver will receive a warning indication in the driving cab on the approach to a:

- colour light signal displaying a single or double yellow (steady or flashing) or a red aspect
- semaphore distant signal displaying a caution indication
- warning indicator provided for some permissible speed reductions
- warning board provided for an ABCL, AOCL or open crossing (OC)
- warning board or emergency indicator for a temporary or emergency speed restriction
- cancelling indicator for an AWS warning which does not apply to the train.

The driver will also receive a warning indication when passing over an AWS depot test magnet.

> ℹ *AWS is not capable of distinguishing between a red, double yellow or single yellow aspect.*

RS522 §1.4.1 *Warning indication*

Clear Indication

The driver will receive a clear indication in the driving cab when approaching:

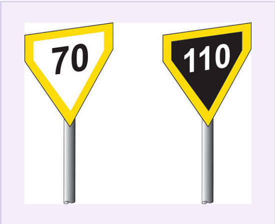
- a colour light signal showing a green aspect, or
- a semaphore distant signal displaying a clear indication.

> ℹ *The driver does not have to acknowledge a clear indication.*

RS522 §1.4.2 *Clear indication*

1.4.3 Signs Associated with AWS Magnets

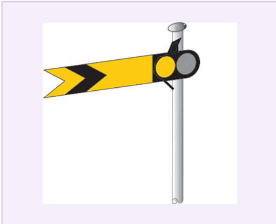
RS522 §1.4.1 lists what gives the driver an AWS warning indication. The images below show what each of those actually looks like on the ground, sourced from RS521 (Signals, Handsignals, Indicators and Signs) and RS522.



Advance Warning of Lower Speed Limit

Provided on the approach to certain permissible speed indicators, warning of a reduction in permissible speed ahead. Black on white = mph; white on black = km/h.

RS521 §7.2 — Warning Indicators



Distant Semaphore Signal (Caution)

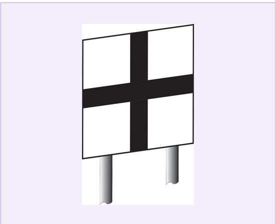
Yellow arm with a fish-tail notch and black chevron, shown at Caution — arm horizontal by day. AWS gives a warning indication when a semaphore distant is at caution.

RS521 §3.1 — Distant Signals

Fixed Distant Signal

A caution indication painted on a board rather than a moving arm — always treated as AT CAUTION. Can be placed before a buffer stop or on the approach to a terminal area.

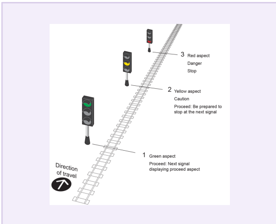
RS521 §3.1 — Distant Signals



Level Crossing Warning Indicator

Marks the approach to an ABCL, AOCL, or open crossing. AWS gives a warning indication on the approach to this board.

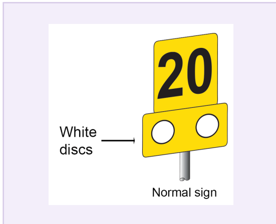
RS521 §6.1 — Level Crossing Signs



Colour Light Signal

Green = proceed. Yellow = caution, be prepared to stop at the next signal. Red = danger, stop. AWS warns for a single/double yellow or red aspect.

RS521 §2.1 — Three-Aspect Signalling



Advance Warning of Temporary Speed Restriction

Placed on the approach to a temporary speed restriction ahead, identified by the two white discs. An AWS magnet is provided on the approach to this board (none in AWS gap areas). A separate speed indicator (number only, no discs) marks the actual start of the restriction.

RS521 §8.1 — Temporary Speed Restriction Signs (Warning Boards)

EROS Board (Emergency Indicator)

Used when an emergency speed restriction is imposed. Flashing white lights work at all times, positioned ahead of any other TSR warning signs.

RS521 §8.2 — Emergency Indicator

1.5 Areas Where AWS is Not Provided

In some AWS fitted areas, AWS equipment is not provided throughout. These areas are identified with the following signs:

Sign	Meaning
Start of AWS gap / End of AWS gap	Where AWS is not provided at a station on a line equipped with AWS.
Start of relevant section / End of the section (normal arrangements resume)	Where AWS is not provided in the wrong direction on a bi-directional line. (If a wrong-direction movement approaches a temporary or emergency speed restriction, AWS will be provided.)

RS522 §1.5 ***Areas where AWS is not provided***

What these signs actually look like:

Start of AWS Gap
Red cross through "AWS", circular board.

End of AWS Gap
Plain "AWS", square board.

Start of Relevant Section
Red cross through "AWS", diamond board (bi-directional line).

End of the Section
Plain "AWS", diamond board (bi-directional line).

1.6 AWS Suppression and AWS Cancelling Indicators

On single and bi-directional lines, the AWS magnet is normally suppressed for movements for which it does not apply and the AWS will not operate.

However, where the AWS magnet is not suppressed, a cancelling indicator is provided to advise the driver that the AWS warning indication does not apply to trains travelling in that direction.

Two types of cancelling indicator signs are used:

- Where the AWS magnet is permanently installed.
- Where the AWS magnet is provided in connection with a temporary or emergency speed restriction.

What these signs actually look like:

Permanently Installed Magnet
Blue and white saltire (X) cross board.

Temporary / Emergency Speed Restriction
Yellow and black saltire (X) cross board, single or bi-directional line.

> *i The cancelling indicator is normally positioned 180 metres (approximately 200 yards) after passing over the AWS magnet, minimum of 45 metres (approximately 50 yards).*

RS522 §1.6 ***AWS suppression and cancelling indicators***

1.7 List of Fault Codes to be Reported

It is important that faults and failures of the AWS or TPWS equipment are reported fully and promptly. This prevents important performance data being lost, allows the signaller to warn following drivers of defective equipment at a specific location, and enables defective on-train equipment to be investigated without delay.

Required Indication	Actual Indication	Fault Code
Clear (Bell)	Horn & Bell	1
	Horn instead of Bell	2
	None	3
Warning (Horn)	Bell & Horn	4
	Bell instead of Horn	5
	Brake without Horn	6
	None	7
	Indicator did not change to Yellow and Black	7a
None	Horn	8
	Bell	9
	Unable to cancel	10
	Indicator did not change to all black	11
	AWS failed to arm	12
	AWS failed to disarm	13
	ATP/TVM failed to arm	14
	ATP/TVM failed to disarm	15
	TPWS failed to activate	16
	TPWS operated when not required	17

Code 7a is not a fault if it occurs immediately after cancelling the AWS indication received when setting a driving cab into service.

i Codes 16 and 17 relate to TPWS specifically, not AWS — included here as they appear in the same table in RS522 §3.

RS522 §3 ***Failures and irregularities — List of fault codes to be reported***

Question & Answer — AWS

All questions and answers are based solely on RS522 Issue 3 (December 2015) — AWS section.

Q1. What does AWS stand for and what is its purpose?

Answer: AWS stands for Automatic Warning System. Its original purpose was to provide the driver with an audible and visual indication of whether the distant signal was clear or at caution. Should the driver fail to respond to a warning indication, an emergency brake application will be initiated. (RS522 §1.1.2)

Q2. How far from a signal is the AWS magnet normally positioned?

Answer: 180 metres (approximately 200 yards) on the approach to a signal. It may be at a greater distance on high-speed lines or at a lesser distance on lower speed and platform lines. (RS522 §1.1.3)

Q3. What is an 'AWS gap area'?

Answer: At through stations where the permitted speed is 30 mph or less and the layout is complex, AWS track equipment need not be provided. Where this occurs, these are called AWS gap areas. (RS522 §1.1.3)

Q4. Are AWS magnets provided at semaphore stop signals?

Answer: No. AWS magnets are not provided at semaphore stop signals. Where a distant signal is mounted on the same post as a semaphore stop signal, AWS is provided for the distant signal. (RS522 §1.1.3)

Q5. What is a 'Morpeth magnet'?

Answer: Where a reduction in permissible speed is provided with a warning indicator (approach speed 60 mph or more, reduction at least one third), an AWS permanent magnet is provided 180 metres on the approach to the warning indicator. These are sometimes referred to as 'Morpeth magnets'. (RS522 §1.1.3)

Q6. What are the five types of AWS track magnet and what does each one do?

Answer: 1. Permanent magnet — triggers warning if no energised electromagnet follows. 2. Electromagnet — when energised, gives a clear indication. 3. Suppressor magnet — suppresses permanent magnets not applicable to a movement. 4. Depot test magnet — tests AWS at depot exits. 5. Portable magnet — used on approach to temporary/emergency speed restrictions. (RS522 §1.2)

Q7. What does the black (fully black) AWS visual indication mean?

Answer: It advises the driver that the associated signal is showing a green aspect or 'all clear'. It also advises that the audible warning has NOT been acknowledged — if not acknowledged, brakes will be applied. (RS522 §1.3)

Q8. What does the yellow and black AWS visual indication mean?

Answer: It advises the driver that a warning indication has been acknowledged. (RS522 §1.3)

Q9. How long does the driver have to acknowledge an AWS audible warning?

Answer: Within two to three seconds. If not acknowledged, an emergency brake application will occur. (RS522 §1.3)

Q10. What does the AWS isolation/fault indicator show when flashing?

Answer: A fault has been detected in the train AWS equipment. (RS522 §1.3)

Q11. What does the AWS isolation/fault indicator show when on steady?

Answer: The train AWS equipment has been isolated. (RS522 §1.3)

Q12. On what aspects/indicators does the driver receive an AWS WARNING indication?

Answer: Single or double yellow (steady or flashing) or red aspect; semaphore distant at caution; warning indicator for permissible speed reductions; ABCL/AOCL/OC warning boards; TSR/ESR warning boards or emergency indicators; cancelling indicators; and depot test magnets. (RS522 §1.4.1)

Q13. Can AWS distinguish between a red aspect, a double yellow, and a single yellow?

Answer: No. AWS is not capable of distinguishing between a red, double yellow or single yellow aspect. (RS522 §1.4.1)

Q14. Does the driver need to acknowledge an AWS clear indication?

Answer: No. The driver does not have to acknowledge a clear indication. (RS522 §1.4.2)

Q15. What signs mark the start and end of an AWS gap area at a station?

Answer: 'Start of AWS gap' and 'End of AWS gap'. (RS522 §1.5)

Q16. What is the purpose of the AWS cancelling indicator?

Answer: Where the AWS magnet is not suppressed, a cancelling indicator advises the driver that the AWS warning indication does not apply to trains travelling in that direction. (RS522 §1.6)

Q17. Where is the cancelling indicator normally positioned?

Answer: Normally 180 metres (approximately 200 yards) after passing over the AWS magnet, with a minimum of 45 metres (approximately 50 yards). (RS522 §1.6)

Q18. Does AWS relieve the driver of the responsibility to observe lineside signals?

Answer: No. AWS does not relieve the driver of the responsibility of observing and obeying lineside signals and indicators. (RS522 §1.1.2)